Year 10 Science
Semester 1 Examination 2018 - Revision

Astronomy

1. Complete the following table.

TERM	DEFINITION
light year	
constellation	
galaxy	
nebula	
supernova	
red giant	
black hole	
sunspot	
corona	
fusion	

2. Light travels at 300,000 km/sec. A star is 97.6 light years away. We can only travel at about 15 - 20 km/second with our best rockets.

   (a) With our current technology, are we likely to be able to travel to this star in the near future? Give a few reasons for your answer.

   YES / NO (Circle your answer.)

   Reasons:
   -
   -
   -

(b) Suggest ***one way*** we might be able to overcome the time factor of travelling to other galaxies, given it could be hundreds of years.

3. (a) What is the major element found in the Sun? ______________

(b) What happens to this element so that the Sun produces its energy?

(c) The Sun loses 5 million tonnes of mass per second. According to Einstein's equation ($E = mc^2$), what does this mass become?

4. How is the surface temperature of a star related to its colour?

TEMPERATURE	**COLOUR**
	blue
	red
	yellow

5. (a) About how old is our Sun (in millions of years)? ______________ million years

(b) How much longer will it "live"? ______________ million years

6. (a) Draw a flow diagram (with arrows) to show the stages of our Sun's remaining life as it runs out of fuel.

Sun → → → →

(b) Do similar flow diagrams for the following types of stars.

Huge star → → →

Giant star → → →

(c) The difference between a ***nova*** and a ***supernova*** is the size of the explosion that takes place. What happens to a star to cause it to explode?

7. (a) Describe the size of the gravity near a black hole.

(b) Why does a black hole appear black?

8. Complete the following matrix to compare the Big Bang and Steady State Theories, using ticks to show which statements apply to one, both or neither theory.

Statement	Big bang theory	Steady state theory
The universe has no beginning.		
The universe began with a single point called singularity.		
The universe is expanding.		
The universe always looks the same.		
The red shift in the spectrum of visible light coming from stars and galaxies provides evidence for the theory.		
New stars and galaxies are created to replace those that move away due to expansion of the universe.		
This theory explains the amount of helium in the universe.		
This theory is supported by the measurement of the current temperature of the universe (about −270 °C).		
This theory was first supported by a Catholic priest.		
The theory will never be proven incorrect.		
An end was put to this theory in 1965.		

9. Why are the constellations we see now so different from the way they were many centuries ago?

10. How have scientists gained their knowledge of the life and death of stars if the processes involved take millions of years to occur?

11. What is the difference between a neutron star and a black hole?

12. What makes stars group together to form galaxies?

13. The Doppler effect is most commonly associated with the changing pitch of a sound as its source moves past you. For example, the pitch of the noise made by a speeding train increases as it approaches you and decreases as it moves away from you. Explain how the Doppler effect is relevant to the study of the universe.

14. What is cosmic microwave background radiation and why does it exist?

Human Biology

1. (a) What are the male and female gametes in animals called?

   (b) What are the ***genotypes*** for a male and a female?

2. DNA is thought to be a twisted ladder structure called a double helix. There are a series of bases located between the two strands of the helix.

   (a) Name the four bases.

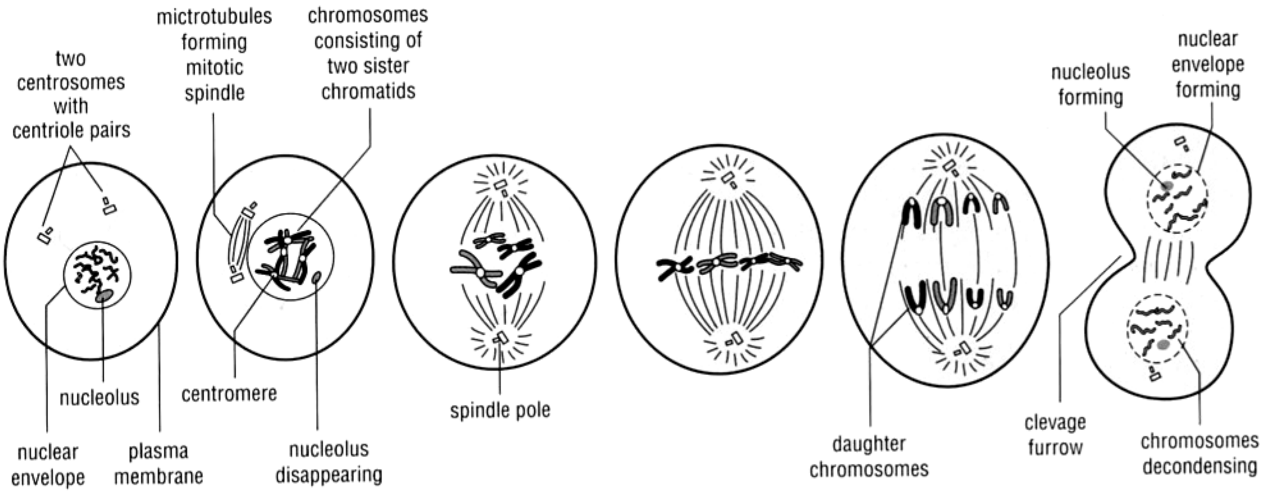
   (b) Which bases are paired together?

   (c) What are the bases made of?

3. Define the following terms.

allele	
pure-bred	
hybrid	
homozygous	
heterozygous	
genotype	
phenotype	
sex-linked	
dominant	
recessive	
co-dominant	
incomplete dominance	
haploid	
gene	

4. Below is a diagram showing a cell division process.



(a) Is this meiosis or mitosis?

(b) Describe what is happening at each stage to the chromosomes.

(c) If possible, name each stage.

(d) In which cells does this process occur?

(e) Draw a diagram to show the stages in the process ***not*** shown above.

5. (a) Why do cells undergo mitosis?

(b) Why do cells undergo meiosis?

(c) In what way do the cells produced by these two processes differ?

6. Polled cattle do not have horns. This is a dominant phenotype arising from the action of an autosomal gene **P**. Horned cattle are therefore **pp** homozygotes.

(a) If a polled bull and a polled cow have a horned calf, what are the genotypes of the parents?

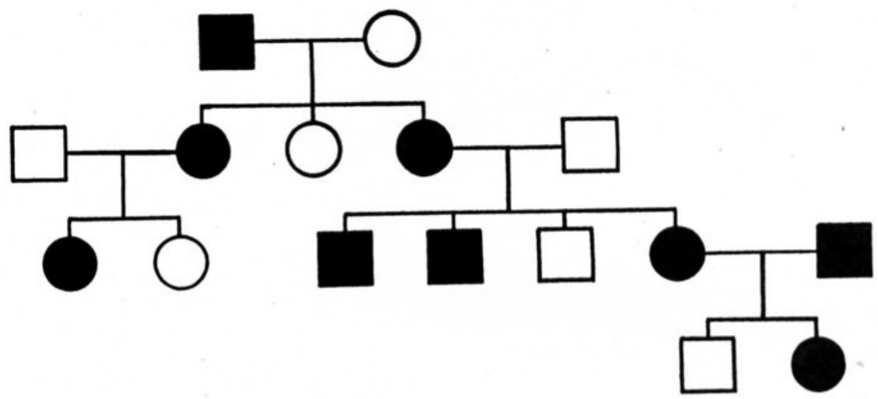
(b) If the same bull was bred with many horned cows, what would the proportions of polled and horned offspring overall be? ***Show full working***.

7. In Andalusian chickens, the black characteristic is ***incompletely dominant*** to the white-splashed characteristic. The heterozygous (hybrid) condition produces blue chickens.

   (a) How is incomplete dominance different to ***co-dominance***?

   (b) Determine the genotype and phenotype of the F1 generation from a pure-bred black Andalusian crossed with a blue Andalusian.

8. The following pedigree shows the occurrence of no ear lobes in a family.



   (a) Is the characteristic dominant or recessive (according to the pedigree)?

   (b) Explain why you made that determination.

   (c) Determine the genotype of every individual in the pedigree.
   (d) How many females have no ear lobes?

9. Describe, with a simple diagram, the first stages of ***meiosis***. Label what is happening at each stage. In your diagrams, use 4 chromosomes as the normal number.

10. (a) How many chromosomes are found in a human skin cell?

(b) How many chromosomes are found in the reproductive cells?

(c) Name the process used to make reproductive cells?

11. (a) What is the difference between the male and female sex chromosome pair?

(b) Who determines the sex of a child - the mother or father? Explain.

(c) Show, by punnet square, why half of any particular population would be expected to be female.

(d) A family has had five boys born into it, and the mother is pregnant with the sixth child. What is the chance that the child will be a girl? Explain why.

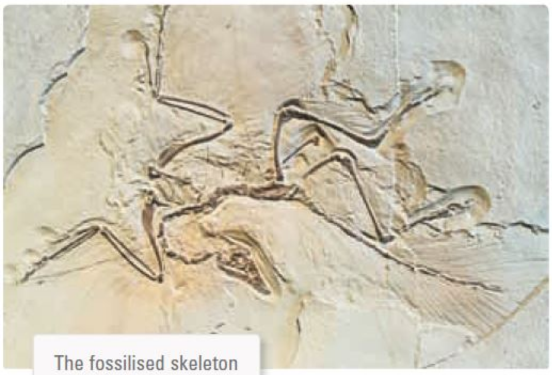
12. (a) Define the term ‘adaptation’.

(b) Explain how each of the following examples are adaptations for survival.

(i) Desert marsupials excrete small quantities of very concentrated urine.

(ii) Polar bears have a thick fur coat.

13. The photo below shows the 150 million year old fossilised remains of an *Archaeopteryx*.



The fossilised skeleton of *Archaeopteryx*

(a) Identify the features that survived the fossilisation process.

(b) Explain why these features survived whereas others did not.

(c) Explain the scientific importance of this fossil.

14. The absolute ages of recent carbonaceous fossils can be determined using radiocarbon dating. The half-life of carbon-14 (a radioactive isotope) is 5730 years. The following table shows how the percentage of C-14 present in a fossil decreases over time.

Time (years)	0	5730	11 460	17 190	22 920	28 650
% C-14 remaining	100	50		12.5	6.25	3.125

(a) Complete the table.

(b) Plot a line graph of this data on the grid below.

(c) The C-14 content of a fossil was found to have decreased by 36% since the organism died. Determine the approximate age of the fossil.

(d) A fossil is approximately 10 000 years old. What percentage of C-14 still remains?

(e) Explain why it is difficult to date fossils that are more than 60 000 years old using radiocarbon dating.